



Pharm*Achieve*

ELECTROLYTE DISORDERS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACEI	Angiotensin Converting Enzyme Inhibitor	MSK	Musculoskeletal
Al³⁺	Aluminum	NSAID	Non-Steroidal Anti-Inflammatory Drug
AG	Anion Gap	NS	Normal Saline
ARB	Angiotensin II Receptor Blocker	PO	By Mouth
CKD	Chronic Kidney Disease	PTH	Parathyroid Hormone
CMP	Comprehensive Metabolic Panel	PPI	Proton Pump Inhibitor
CNS	Central Nervous System	QT	QT Wave Interval
D5W	Dextrose 5% in Water	SIADH	Syndrome of Inappropriate Antidiuretic Hormone
ECG	Electrocardiogram	SNRI	Serotonin-Norepinephrine Reuptake Inhibitor
HyperPTH	Hyperparathyroidism	SSRI	Selective Serotonin Reuptake Inhibitor
HypoPTH	Hypoparathyroidism	TCA	Tricyclic Antidepressant
IV	Intravenous	TdP	Torsades de Pointes

DEFINITIONS

	Hypo-	Insufficient physiologic concentration
	Hyper-	Excessive physiologic concentration
Na^+	-natremia	Referring to sodium in blood
K^+	-kalemia	Referring to potassium in blood
Ca^{2+}	-calcemia	Referring to calcium in blood
PO_4^{3-}	-phosphatemia	Referring to phosphorus (phosphate) in blood

DIAGNOSIS OF ELECTROLYTE DISORDERS

- Laboratory analysis of electrolyte levels
 - Normal values vary by evaluating laboratory
- ECG
- Vital signs
- Medical history (e.g. chronic kidney disease)
- Physical assessment (e.g. malnourishment)
- Social history (e.g. alcoholism)
- Musculoskeletal assessment (e.g. reflexes)
- Neurological status
- Other symptoms that clinically correlate to altered electrolytes
 - Fatigue, muscle cramps, nausea, vomiting, etc.
 - Important to look at the PATIENT, not just the numbers!

GOALS OF THERAPY

- ✓ Normalize electrolyte levels
- ✓ Treat underlying cause
- ✓ Prevent recurrence
- ✓ Prevent and treat clinical consequences

SODIUM (Na⁺)

Normal Na⁺ range:
136 mEq/L – 146 mEq/L

Extracellular cation
Responsible for tonicity and osmotic pressure
Linked to volume status of the body

HYPONatremia
(<136 mEq/L)

- Nausea/vomiting
- Confusion
- Headaches
- Muscle cramps
- Weakness
- Lethargy
- Seizures
- Coma
- Death

- Extreme thirst
- Lethargy
- Weakness
- Confusion
- Muscle irritability/spasms
- Seizures
- Coma
- Death

HYPERnatremia
(>146 mEq/L)

HYPONATREMIA

Causes

- Must also assess patient's volume status and osmolality to determine etiology
- Drug causes of Hypovolemic hyponatremia: thiazide **diuretics**, loop **diuretics**, osmotic **diuretics** (e.g. glucose, mannitol, urea)
- Drug cause of SIADH: antineoplastic agents (e.g. cyclophosphamide, vincristine), antipsychotics (e.g. haloperidol), carbamazepine, desmopressin, **NSAIDs**, opioids, **SSRIs/SNRIs**, TCAs, **lithium**, oxytocin

Treatment

- Treat underlying cause
 - Remove any offending agents
- Hypertonic saline, normal saline, Ringer's lactate
- Water restriction for isovolemic hyponatremia (e.g. SIADH)
- Acute (<48 hours): replete levels immediately
- Chronic (>48 hours): replete levels gradually ≤ 10 mEq/L/24 hours
 - Faster repletion can increase risk of central pontine myelinolysis

HYPERNATREMIA

Causes

- Loss of fluids
 - Diarrhea, vomiting, nasogastric suctioning, burns, open wounds, etc.
- Diabetes insipidus
- Hyperaldosteronism
- Iatrogenic causes including hypertonic saline, sodium bicarbonate solutions

Treatment

- Treat underlying cause
- Volume resuscitation – Free water provision (D5W or 0.45% NS, etc.)
- Acute (<48 hours) situations should be corrected rapidly
- Chronic hypernatremia should be corrected gradually to avoid damaging the CNS

POTASSIUM (K⁺)

Normal K⁺ Range:
3.5-5.1 mEq/L

Intracellular cation
Important for cellular metabolism and regulation of action potentials across cell membranes

HYPOkalemia
(<3.5 mEq/L)

- Muscle weakness and wasting
- QTc prolongation
- Arrhythmia (TdP)

- Muscle weakness
- Ventricular-tachy arrhythmias
- ECG might show peaked T waves and prolonged QRS complex
- Acidosis

HYPERkalemia
(>5.1 mEq/L)

HYPOKALEMIA

Causes

- Drug related causes: insulin, amphotericin B, loop and thiazide diuretics, beta agonists
- Fluid loss from causes such as vomiting, diarrhea, excessive sweating, etc.
- May be inaccurately diagnosed in patients with diabetic ketoacidosis – why?

Treatment

- Potassium supplementation (PO or IV)
 - IV potassium should never be given as an IV push injection
 - Continuous cardiac monitoring is required when replenishing more than 10mEq/hour
- Can expect potassium to raise approximately 0.1 mEq/L per each 10 mEq given

HYPERKALEMIA

Causes

- Drug causes: Potassium-sparing diuretics, ACE-i/ARB, trimethoprim, NSAIDs
- Chronic Kidney Disease
- Diet very high in potassium

Treatment

- Treat underlying cause and remove exacerbating agents
- Drugs that push potassium into cells: insulin, sodium bicarbonate, beta agonists, loop and thiazide diuretics
- Potassium binder: sodium polystyrene sulfonate
- Dialysis in severe cases
- Calcium gluconate or calcium chloride may be given to prevent cardiac arrhythmia

MAGNESIUM (Mg^{2+})

Normal Mg^{2+} Range:
0.74-1.03 mmol/L

Intracellular cation

Cofactor for potassium absorption

Involved in cardiac, nerve, bone, and muscle functioning

HYPOMagnesemia
(<0.74 mmol/L)

- Muscle weakness
- QTc prolongation
- Arrhythmia (TdP)
- Hyperreflexia/tetany
- Seizures

- Hypotension
- Arrhythmias
- ↓ tendon reflexes
- CNS depression
- Facial flushing

HYPERmagnesemia
(>1.03 mmol/L)

HYPOMAGNESIMIA

Causes

- Drug-related causes: diuretics, aminoglycosides, long-term proton pump inhibitor use
- Insufficient intake in the diet
- Chronic alcoholism

Treatment

- Remove any offending agents if possible
- Magnesium supplementation (PO or IV)
- Encourage balanced diet with good sources of magnesium such as dark green, leafy vegetables

HYPERMAGNESEMIA

Causes

- Drug causes: Magnesium-containing antacids and laxatives most commonly
- Chronic Kidney Disease
- Excessive intake is a less common cause

Treatment

- Treat the underlying cause and remove exacerbating agents
- Calcium gluconate or calcium chloride (antagonizes the effects of magnesium)
- Diuretics
- Hemodialysis in severe cases

CALCIUM (Ca^{2+})

Normal Ca^{2+} Ranges:
Total: 2.12-2.52 mmol/L
Ionized: 1.14-1.28 mmol/L

Mostly stored in bones and teeth

Ionized (free) calcium is what is physiologically active

Important for bone, nerve, muscle, and heart health

HYPOcalcemia
(total < 2.12 mmol/L)

- Tetany
- Spasms
- Prolonged QT
- Heart failure symptoms

- MSK weakness
- Constipation
- Kidney stone formation
- Seizures
- Confusion
- Coma
- Shortened QT

HYPERcalcemia
(total > 2.52 mmol/L)

HYPOCALCEMIA

Causes

- Drug causes: Bisphosphonates (IV >>> PO), denosumab, PPIs, cinacalcet
- Phosphorous-containing products (precipitate formation)
- Hypoparathyroidism*
- Low vitamin D levels

* Although hypocalcemia is more commonly caused by *HYPO*parathyroidism, there are other *less* common causes that are associated with *hyper*parathyroidism, e.g. vitamin D deficiency/resistance, chronic kidney disease, parathyroid hormone resistance

Treatment

- Treat underlying cause and remove exacerbating agents
- Calcium chloride or calcium gluconate
 - Acute hypocalcemia should be treated with IV calcium
 - In chronic hypocalcemia, can use PO options
 - Should include vitamin D or calcitriol with calcium in patients with chronic hypocalcemia

HYPERCALCEMIA

Causes

- Drug causes: thiazide diuretics, lithium, teriparatide
- Hyperparathyroidism
- Diet high in calcium, vitamin D, and/or vitamin A
- Malignancies

Treatment

- Treat underlying cause and remove exacerbating agents
- Hydration
- IV bisphosphonates
- Calcitonin
- Loop diuretics
- Dialysis in severe cases

PHOSPHATE (PO_4^{3-})

Normal Range:
0.81–1.58 mmol/L

Intracellular cation

Closely linked to calcium levels

Important for bone health, cellular processing/energy, and waste excretion

HYPOphosphatemia
(<0.81 mmol/L)

- Generally asymptomatic
- Weakness
- Confusion
- Platelet dysfunction
- Hemolysis
- Respiratory failure

- Usually asymptomatic, but may present with symptoms of low Ca^{2+}
- Tetany
- Confusion
- Vascular calcification

HYPERphosphatemia
(>1.58 mmol/L)

HYPOPHOSPHATEMIA

Causes

- Drug causes: bisphosphonates, glucocorticoids, diuretics, antacids
- Starvation and refeeding syndrome
- Alcoholism
- Hyperparathyroidism
- Low vitamin D

Treatment

- Treat underlying cause and remove exacerbating agents
- Supplementation via sodium phosphorous, potassium phosphate, effervescent phosphate tablets, etc.
- Mild cases may be treated with diet modifications

HYPERPHOSPHATEMIA

Causes

- Drug causes: phosphate-based enemas, drugs with phosphorous as an excipient
- Chronic Kidney Disease
- Tumour Lysis Syndrome
- Hypoparathyroidism
- Low calcium levels
- Excess vitamin D and/or vitamin A

Treatment

- Treat underlying cause and remove exacerbating agents
- Diet restriction (avoiding high phosphate foods like dairy, meat, legumes, etc.)
- Increase hydration
- Phosphate binders such as calcium carbonate, sevelamer, calcium acetate, etc.

BICARBONATE BUFFER SYSTEM

What is the bicarbonate buffer system?

- Mechanism by which the body maintains homeostasis regarding acid-base balance – average physiological pH = 7.4 in the blood (ranges between 7.35 to 7.45)
- The system is a balance of carbonic acid, bicarbonate, and carbon dioxide and water (represented as $\text{H}^+ + \text{HCO}_3^- \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}_2\text{O} + \text{CO}_2$)
 - Carbon dioxide reacts with water to form carbonic acid, which is dissociated to the bicarbonate ion + a hydrogen ion
 - Carbonic anhydrase acts as the catalyst in this reaction and is an important enzyme in the buffer system

What happens if it is disrupted?

- If physiologic pH is unable to be maintained, it may result in an acid-base disorder
 - Acidosis = pH < 7.35
 - Alkalosis = pH > 7.45
- Maintaining a proper pH in the blood is vital for proper metabolic functioning; therefore, any imbalance needs to be addressed immediately

ANION GAP

What is an anion gap?

- The anion gap is the difference between anions and cations in the body's fluids, generally measured in the plasma
- An anion gap allows practitioners to see if there are electrolyte abnormalities in an acid-base disorder, as this will alter the treatment plan
 - Most common anion gap-related disorder is high anion gap metabolic acidosis

How do you calculate an anion gap?

- Using CMP results, calculate as below:
 - $(\text{Na}^+ + \text{K}^+) - (\text{Cl}^- + \text{HCO}_3^-) = \text{AG}$
 - Normal range = 5-12 mmol/L
- Some institutions do not include potassium in the AG calculation

HIGH ANION GAP CAUSES

- Cyanide or carbon monoxide
- Arsenic
- Toluene
- Methanol or metformin
- Uremia
- Diabetic ketoacidosis
- Paraldehyde
- Iron or isoniazid
- Lactate
- Ethylene glycol
- Salicylates

ACIDOSIS

TYPE OF ACIDOSIS	METABOLIC High anion gap	METBOLIC Normal anion gap	RESPIRATORY
Buffer Changes	Increase in H ⁺ ions, generally from lactic acid	Decreased bicarbonate	Increased CO ₂ → increased carbonic acid
Compensatory Mechanisms	- Increased respirations - Intracellular shift of hydrogen ions - Increase bicarbonate via reabsorption in kidney or via buffer system		- Increase bicarbonate - Renal carbonic acid secretion/bicarbonate reabsorption
Causes	"CAT MUDPILES" Focus on medications as a pharmacist	- Loss of fluids - Renal dysfunction - Carbonic anhydrase inhibitors	Failure of ventilation - COPD/asthma - ALS - Myasthenia gravis
Symptoms	- Palpitations - Anxiety - Nausea/vomiting - Abdominal pain		- Headache - Restlessness - Anxiety - Somnolence
Treatment	- Treat underlying cause - IV (severe) or PO (in chronic) bicarbonate		- Treat underlying cause - Bronchodilators

ALKALOSIS

TYPE OF ALKALOSIS	METABOLIC Chloride Responsive	METABOLIC Chloride Resistant	RESPIRATORY
Buffer Changes	Increased bicarbonate or decrease carbonic acid depending on exact cause		Decreased carbon dioxide
Compensatory Mechanism	Hypoventilation to retain carbon dioxide which is then catalyzed to carbonic acid to raise pH		Buffer shift to increased carbon dioxide levels
Causes	• Cystic Fibrosis • Diuretics • Antacids • Vomiting	• Cushing syndrome • Fludrocortisone • Chewing tobacco products	• Salicylate overdose • High altitude • Hyperventilation
Symptoms	• Electrolyte changes • Seizures • Confusion	• Hypoventilation • Tachycardia • Reduced blood flow	• Light-headedness • Sweating • Fatigue
Treatment	• Treat underlying cause • Trial of IV of chloride	• Treat underlying cause • K⁺ or spironolactone	• Treat underlying issue • Brown bag breathing

TAKEAWAY

- Treat the patient, not the number
 - Important to objective and subjective findings for best plan of action
- Beware of treating with agents that contain more than one electrolyte
 - E.g.: If patient has low phosphate but high potassium, avoid using potassium phosphate to correct the abnormal phosphate
- Potassium should NEVER be given via IV push – IV infusion or oral options only
 - Potassium requires cardiac monitoring if infusing more than 10 mEq/hour
- Treatment options involve removing exacerbating agents, monitoring and adjusting hydration status, and
 - Replenishing electrolytes (hypo-)
 - Using other agents and/or dialysis to bind and remove excess electrolytes (hyper-)

CASE 1

NC is a 76-year-old male who presents with lethargy and confusion. NC was at his general practitioner (GP) for his routine checkup and was found to have elevated blood pressure. His physician started him on perindopril-indapamide 4 mg-2.5 mg PO daily 2 weeks ago. Since starting, his blood pressure has been under control, but his family notes that he has been more confused in the past few days, sometimes not recognizing where he is or who his family members are. He seems to be urinating more often, and his mucous membranes are dry. NC has a past medical history of a cerebrovascular event (CVA) in 2014 and hypertension for which he takes clopidogrel 75 mg PO daily, rosuvastatin 10mg PO daily, and acetaminophen 500 mg PRN. He is allergic to Aspirin®. He does not smoke or drink alcohol.

Objective findings:

- BP=154/73 mmHg, HR=77 bpm
- ECG normal
- Na=121 mmol/L
- K=3.8 mmol/L
- Mg=0.76 mmol/L
- Ca=2.2 mmol/L
- SCr=114 umol/L



CASE 1 - QUESTIONS

1. What electrolyte abnormality is most likely to contribute to his symptoms?
 - a) Sodium
 - b) Potassium
 - c) Magnesium
 - d) Calcium
2. Which of Mr. NC's medications can cause changes in the above electrolytes?
 - a) Clopidogrel
 - b) Perindopril
 - c) Indapamide
 - d) Rosuvastatin
3. Which of the following treatment strategies would be most appropriate?
 - a) Hypertonic saline 3% at 50 mL/hour
 - b) Change perindopril-indapamide to perindopril 4 mg PO daily alone, and initiate 0.9% NaCl at 100mL/hour
 - c) Stop indapamide and fluid restrict to 1.5 L per day
 - d) Sodium chloride oral tablets with meals

CASE 2

KC is a 58-year-old male who presents with weakness and fatigue. KC had increasing edema in his legs and mild shortness of breath on exertion 4 weeks ago. He presented to his GP, who started him on furosemide and scheduled him for an echocardiogram. Since then, both symptoms have improved, but he has noticed increasing weakness in the past week. KC has a past medical history of hypertension and chronic obstructive pulmonary disease (COPD) for which he takes atenolol 50 mg PO daily, furosemide 40 mg PO daily, tiotropium 18 mcg inhaled daily, multivitamin 1 tab PO daily, and acetaminophen 500 mg PO QID PRN. He smokes 10 cigarettes/day, drinks 2-3 glasses of alcohol on weekends, and has no allergies to medications.

Objective findings:

- BP=132/85 mmHg, HR=87 bpm
- ECG normal
- Na=137 mmol/L
- K+=2.9 mmol/L
- Mg=0.65 mmol/L
- Ca=2.21 mmol/L
- SCr=98 umol/L



CASE 2 - QUESTIONS

1. What electrolyte abnormality is most likely to contribute to his symptoms?
 - a) Phosphate and calcium
 - b) Potassium and calcium
 - c) Magnesium and phosphate
 - d) Potassium and magnesium
2. Which of KC's medications can cause changes in the above electrolytes?
 - a) Atenolol
 - b) Furosemide
 - c) Tiotropium
 - d) Acetaminophen
3. What treatment is most suitable for KC?
 - a) Sodium polystyrene sulfonate 15g PO x once
 - b) Potassium chloride 10-20 mmol IV once over 1 hour
 - c) Potassium chloride 40 mEq PO Q4H x 2 doses
 - d) Potassium chloride 10-20 mmol IVP once
4. You replenish KC's affected electrolyte and repeat bloodwork, but his level continues to be low. What could be the most likely cause?
 - a) Insufficient replacement
 - b) Oral option is insufficient
 - c) Insufficient cofactor for absorption
 - d) Did not allow long enough for absorption

CASE 3

CA presents with decreased level of consciousness (LOC) and confusion. CA is a 56-year-old female who was recently diagnosed with breast cancer and has started her first round of chemotherapy. Her family noticed that she had been more fatigued and attributed that to the chemo. However, in the last 2 days she has been more confused and this morning she was barely rousable. Her husband called 911 and EMS brought her to the ER. CA has a past medical history of breast cancer, osteoporosis, and hypothyroidism. She takes levothyroxine 88 mcg po daily, doxorubicin and cyclophosphamide IV q21days– chemotherapy, calcium carbonate 500 mg PO bid, and vitamin D₃ 3000 units PO daily. She does not smoke or drink alcohol and she has no allergies to any medications.

Objective findings:

- BP=115/73 mmHg, HR=70 bpm
- Na=136 mmol/L
- K=4.1 mmol/L
- Ca=2.98 mmol/L
- Albumin=32.1 g/L
- PO₄=0.87 mmol/L
- SCr=104 umol/L



CASE 3 - QUESTIONS

1. What electrolyte disturbance is most likely causing her symptoms?
 - a) Sodium
 - b) Potassium
 - c) Calcium
 - d) Phosphate
2. What factors may be contributing to her above-elevated level?
 - a) Breast cancer
 - b) Low albumin
 - c) Vitamin D supplement
 - d) All of the above
3. What therapeutic option would you consider first for Ms. CA?
 - a) NS IV at 200 mL/hr
 - b) Pamidronate 90 mg IV x once
 - c) Dialysis
 - d) Furosemide 20 mg IV x once

CHANGE LOG

June 2025

- Significant changes throughout

November 2025

- No content changes made

January 2026

- Slides 17 & 18: Changed from hyperparathyroidism to *hypoparathyroidism* as a cause of hypocalcemia

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 13: Updated signs/symptoms of hypo- and hypermagnesemia